

Credit Risk Assessment & Loan Default Analysis

SQL | Power BI | Business Intelligence

Portfolio Project

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Role: Data Analyst

Tools Used: SQL • Power BI • Microsoft Excel

Executive Summary

This project analyses a loan application dataset to identify the major factors influencing loan approval decisions and credit risk.

Using SQL for data extraction and Power BI for data transformation, dashboard development and visualization, the analysis provides insights into applicant demographics, employment status, income, credit score, years of experience and geographical distribution.

The dashboard reveals that although 5,000 loan applications were received, only 1,151 applications (23%) were approved while 3,849 (77%) were rejected. This indicates that the lending institution applies a highly selective credit approval process designed to minimise loan default risk.

The analysis demonstrates how data analytics enables financial institutions to make objective lending decisions, reduce non-performing loans, improve profitability and optimise portfolio performance.

Project Objectives

The objectives of this project were to:

- *Analyse customer loan applications.
- *Identify key drivers of loan approval.
- *Understand potential credit risk.
- *Evaluate customer characteristics.
- *Discover approval trends across demographic groups.
- *Produce an interactive dashboard for management decision-making.

Tools Used

Tool	Purpose
SQL	Data extraction
Power BI	Transformation, Filtering and Dashboard development
Power Query	Data cleaning
DAX	KPI calculations
Excel	Initial validation

Business Questions Answered

The project answers several important lending questions:

- *What percentage of applications are approved?
- *Which employment group receives the most approvals?
- *Does age influence approval?
- *Does credit score significantly impact lending?
- *Which city has the highest loan demand?
- *How does work experience affect approval?
- *Which customer profiles present higher lending risk?

The screenshot shows a SQL query editor with a toolbar at the top. The query is as follows:

```
2 • SELECT
3     Age,
4     Income,
5     LoanAmount,
6     CreditScore,
7     YearsExperience,
8     Gender,
9     Education,
10    City,
11    EmploymentType,
12    LoanApproved
13
14    FROM `loan_database`.`Clients_loan_records`;
15
```

Below the query editor is an "Output" section with a dropdown menu set to "Action Output". The output table has columns for "#", "Time", "Action", and "Message".

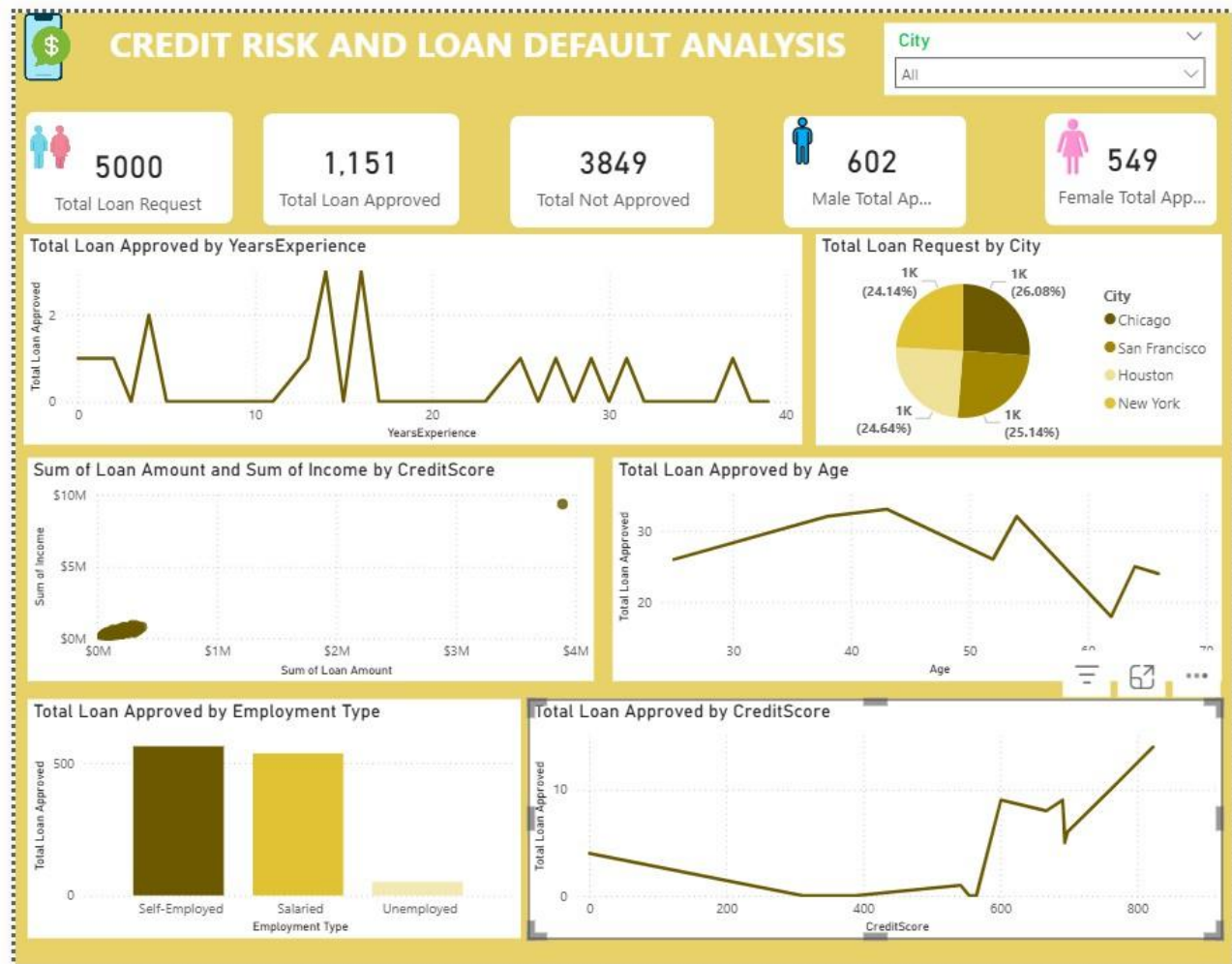
The screenshot shows a data analysis application with a dark theme. The top bar includes a search bar and a "Sign in" button. The main toolbar contains various icons for data manipulation and analysis. The central area displays a table of loan records with the following columns: Age, Income, LoanAmount, CreditScore, YearsExperience, Gender, Education, City, EmploymentType, LoanApproved, and Loan Categorization. The table contains 20 rows of data. On the right side, there is a "Data" panel with a search bar and a list of actions for the "loan risk prediction dataset", including "New measure", "New column", "New quick measure", "Refresh", "Edit query", "Incremental refresh", "Manage aggregations", "Copy table", "Rename", "Delete from model", "Hide in report view", "Unhide all", "Collapse all", and "Expand all".

Age	Income	LoanAmount	CreditScore	YearsExperience	Gender	Education	City	EmploymentType	LoanApproved	Loan Categorization
61	\$39,521	\$13,080	560		2 Female	Masters	Chicago	Unemployed	0	Not Approved
21	\$51,494	\$30,311	360		1 Male	Masters	Chicago	Unemployed	0	Not Approved
42	\$55,385	\$19,293	327		10 Male	Masters	Chicago	Unemployed	0	Not Approved
41	\$58,998	\$20,489	0		27 Female	Masters	Chicago	Unemployed	0	Not Approved
25	\$39,523	\$9,083	609		26 Male	Masters	Chicago	Unemployed	0	Not Approved
52	\$56,676	\$34,279	413		28 Female	Masters	Chicago	Unemployed	0	Not Approved
49	\$27,936	\$1,138	0		35 Male	Masters	Chicago	Unemployed	0	Not Approved
50	\$31,743	\$22,539	751		28 Male	Masters	Chicago	Unemployed	0	Not Approved
57	\$76,532	\$19,128	734		32 Female	Masters	Chicago	Unemployed	0	Not Approved
65	\$66,694	\$28,114	0		14 Female	Masters	Chicago	Unemployed	0	Not Approved
23	\$74,617	\$25,507	849		34 Male	Masters	Chicago	Unemployed	0	Not Approved
18	\$58,084	\$27,744	329		6 Female	Masters	Chicago	Unemployed	0	Not Approved
67	\$40,776	\$9,893	334		24 Female	Masters	Chicago	Unemployed	0	Not Approved
41	\$62,783	\$30,710	394		35 Female	Masters	Chicago	Unemployed	0	Not Approved
34	\$42,221	\$19,537	824		39 Male	Masters	Chicago	Unemployed	0	Not Approved
52	\$23,196	\$20,556	570		38 Male	Masters	Chicago	Unemployed	0	Not Approved
62	\$23,531	\$16,756	849		38 Female	Masters	Chicago	Unemployed	0	Not Approved
67	\$33,010	\$7,072	303		13 Male	Masters	Chicago	Unemployed	0	Not Approved

Table: loan_risk_prediction_dataset (5,000 rows)

Dashboard KPIs

KPI	Value
Total Loan Requests	5,000
Loans Approved	1,151
Loans Rejected	3,849
Approval Rate	23%
Rejection Rate	77%
Male Approvals	602
Female Approvals	549



Key Business Insights

1. Low Overall Approval Rate

Only 23% of loan applications were approved.

Business Interpretation

The organisation appears to follow a conservative lending policy.

This significantly reduces the probability of future loan defaults.

While stricter approval criteria reduce bad debts, they may also limit business growth by rejecting potentially profitable customers.

2. Employment Status is a Major Predictor

Self-employed and salaried applicants account for almost all approved loans.
Unemployed applicants have very few approvals.

Business Meaning

Employment provides evidence of income stability.
Applicants with regular income are considered lower credit risks because they are more likely to make scheduled repayments.

3. Credit Score Strongly Influences Approval

The dashboard clearly shows approvals increasing as credit score rises.
Customers with stronger credit histories receive significantly more approvals.

Business Meaning

Credit score remains one of the strongest predictors of repayment behaviour.
Higher scores generally indicate:
*lower default probability
*responsible borrowing history
*stronger financial discipline

4. Years of Experience

Applicants with more years of employment generally receive more approvals.

Business Interpretation

Long employment history suggests:
*financial stability
*consistent income
*lower repayment risk

This aligns with standard banking credit assessment models.

5. Age Distribution

Loan approvals are concentrated among middle-aged applicants.
Very young and older applicants receive fewer approvals.

Possible Explanation

Middle-aged customers often:
*have established careers
*stable income
*stronger credit history
*higher repayment capacity

6. Geographic Loan Demand

Loan requests are distributed fairly evenly across:
*Chicago
*New York
*Houston
*San Francisco
No city dominates total demand.

Business Insight

Demand appears geographically diversified.

This reduces concentration risk within the lender's portfolio.

7. Gender Distribution

Loan approvals are almost evenly distributed:

*Male: 602

*Female: 549

Business Interpretation

The lending process appears relatively balanced across gender, suggesting approval decisions are primarily driven by financial risk indicators rather than demographic characteristics.

Credit Risk Assessment

The dashboard highlights several indicators commonly used in credit risk modelling:

Low Risk Customers

*High credit scores

*Stable employment

*Higher income

*Longer work experience

*Moderate loan amounts

These customers are more likely to repay loans on time.

High Risk Customers

Applicants showing one or more of the following characteristics may present higher credit risk:

*Low credit scores

*Unemployment

*Limited work experience

*High loan amounts relative to income

*Weak repayment capacity

Such customers are more likely to default.

Business Impact

Effective credit risk assessment produces several business benefits:

Reduced Bad Debt

Approving financially stronger applicants reduces loan losses.

Improved Cash Flow

Timely repayments increase available cash for issuing new loans.

Higher Profitability

Lower default rates improve net interest income and reduce impairment expenses.

Better Capital Allocation

Resources can be directed toward low-risk, profitable borrowers.

Stronger Regulatory Compliance

Data-driven lending supports responsible lending practices expected by financial regulators.

Financial Statement Impact

Income Statement

Improved lending decisions can lead to:

- *higher interest income
- *fewer loan impairment charges
- *improved operating profit
- *higher net income

Balance Sheet

Assets

Performing loans remain valuable assets because they continue generating interest income.

Poor-quality loans eventually become impaired assets.

Loan Loss Provision

Higher default risk requires larger provisions.

Better credit screening reduces these provisions.

Cash

Consistent repayments improve operating cash flow.

Shareholders' Equity

Higher retained earnings strengthen shareholders' equity over time.

Risk Recommendations

The lending institution should consider:

- *Using predictive credit scoring models.
- *Implementing machine learning for default prediction.
- *Continuously monitoring borrower behaviour.
- *Get information from customers about home ownership and analyse it. Homeowners generally presented lower credit risk than non-homeowners.
- *Performing affordability assessments before approval.
- *Reviewing lending policies regularly.

SQL Example Used

```
SELECT
    Age,
    Income,
    LoanAmount,
    CreditScore,
    YearsExperience,
    Gender,
    Education,
    City,
    EmploymentType,
    LoanApproved
FROM `loan_database`.`Clients_loan_records`;
```

Example Power BI DAX Measures

Total Applications =
COUNTROWS(Loans)

Total Approved =
CALCULATE(
COUNTROWS(Loans),
Loans[LoanStatus]="Approved")

Approval Rate =
DIVIDE([Total Approved],
[Total Applications],0)

Key Skills Demonstrated

- *SQL querying
- *Data extraction
- *Data cleaning
- *Data modelling
- *Power BI dashboard development
- *DAX calculations
- *KPI reporting
- *Credit risk analysis
- *Financial analysis
- *Business intelligence
- *Data storytelling
- *Executive reporting

Conclusion

This analysis demonstrates how data analytics supports better credit risk management and evidence-based lending decisions. Although only 23% of loan applications were approved, the conservative approval strategy helps protect the institution from excessive credit losses by prioritising applicants with stronger financial profiles, such as higher credit scores, stable employment, and greater work experience.

From a business perspective, the findings show that robust credit assessment improves portfolio quality, reduces loan defaults, strengthens cash flow, and enhances long-term profitability. By combining SQL for data extraction with Power BI for interactive visualisation, this project illustrates the practical application of data analytics in financial services and highlights how data-driven insights can support strategic decision-making.

For recruiters in Ireland, this portfolio demonstrates not only technical proficiency in SQL, Power BI, DAX, and data visualisation but also the ability to translate data into meaningful business recommendations—an essential skill for a Data Analyst working in banking, financial services, insurance, or risk management.